Data Science for Biologists: A Hands-On Introduction with R and Bioconductor

Sean Davis Lori Shepherd Kern Martin Morgan

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Summary

The RBioc Book is an open-access, executable textbook that takes a reader from their first line of R to the analysis of real genomic data with Bioconductor (Huber et al. 2015). It is written for biologists and biomedical trainees who know the biology and have been introduced to programming, statistics, and computation but want a more formal and thorough treatment of concepts and skills. The book also includes biology primer, for computational readers approaching biological data for the first time. The book is built as a Quarto book (Allaire et al. 2024) in which every chapter executes its own R (R Core Team 2024) code at render time, so the prose, the code, and the output the reader sees are always in sync.

Unlike many other online materials for teaching R and statistics, the contents of the book are meant to explore contents, provide the "why" for material, and to shore up statistical and data science reasoning through clear descriptions, callouts, and explanatory figures. The goal is to allow learners to ingest material at their own paces and to allow instructors time to engage with course participants in the context of a synchronous or asynchronous teaching environment. The book is published online, versioned, and archived with a citable DOI.

Statement of need

Biologists increasingly need to understand statistical and data science principles, but many generic "intro to R" courses show HOW to perform analysis but don't motivate with concept and rationale. Yet, most biomedical researchers have never taken a formal course in statistics or data science. Yet, biomedical researchers are expected to drive analyses of their own data, and to do so in a reproducible manner. This book is meant for adult learners who want to address well documented gaps (Drnevich et al. 2025) in their training and to gain a more formal understanding of the principles of data science and statistics and to equip them with the skills to perform reproducible analyses of their own data.

The RBioc Book is designed to close that gap with one continuous path. Rather than teaching programming in the abstract and leaving the biology for "later," it carries biological framing from the first chapters and builds the conceptual scaffolding — data structures, the tidyverse, statistical reasoning — that the genomics chapters later depend on. Because the book is fully executable and openly licensed, an instructor can adopt a chapter as a lab, fork it for a local course, or assign it for self-paced study, and a learner can reproduce every result by running the same code. It is feature-complete and has been used, in whole and in part, in formal courses and by a broad online readership (see *Evidence of use*).

Instructional design and content

The book is organized around adult-learning principles (Knowles et al. 2015): each chapter opens with explicit learning objectives, motivates concepts before formalizing them, and pairs worked examples with exercises and solutions. A consistent callout taxonomy (objectives, notes, warnings, exercises) gives the reader predictable signposts throughout. The narrative is intentionally cumulative — early chapters earn the vocabulary and data structures that the statistics, machine-learning, and genomics chapters reuse — so that by the time a reader reaches `SummarizedExperiment` or single-cell analysis, the container abstractions feel like an extension of skills they already have rather than new machinery.

The genomics material is taught against real, public datasets and the Bioconductor classes practitioners actually use, so the skills transfer directly to the reader's own analyses. A biology primer lowers the barrier for readers without a biology background, making the book usable from either direction. All materials provide substantial context and rationale for the analyses, so that readers understand not just how to perform an but why.

Evidence of use

The RBioc Book has been taught in four formal offerings: the Cold Spring Harbor Laboratory data-analysis course (2025 and 2026) and the BigCARE program (2025 and 2026), each with approximately 25 participants per session — roughly 100 trainees over two years. Beyond the classroom, the online edition has reached a broad audience: between May 2024 and June 2026, the site recorded 7,513 page views across all the chapters and 876 readers, with substantial per-chapter engagement (for example, the single-cell setup chapter averaged over seven minutes of active engagement per reader), indicating that readers work through the material rather than skim it. Authorship and per-chapter contributions reflect both the primary author and members of the Bioconductor Core Team, and the book continues to be developed in the open with community contributions.

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